WORKED SOLUTION

STEP III 2000, Question 14

Probability and Statistics • Joint distribution

STEP III 2000, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q14

The key idea

Fill in the missing fourth probability, then write out $\mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y)$ term by term. Everything collects into a single factor $a - pq$ times something that cannot vanish, and $a = pq$ is exactly the statement that the top-left cell factorises. Once one cell factorises in a 2×2 table, the margins force the other three to as well. With three values there is more room, and a symmetric table breaks the link.

Solution

The identity

The four probabilities sum to 1, so

$$\mathrm{P}(X = x_2,\, Y = y_2) = 1 - a - (q - a) - (p - a) = 1 - p - q + a.$$

Adding along rows and columns gives the marginal distributions:

$$\mathrm{P}(X = x_1) = a + (q - a) = q, \qquad \mathrm{P}(Y = y_1) = a + (p - a) = p,$$

so $\mathrm{P}(X = x_2) = 1 - q$ and $\mathrm{P}(Y = y_2) = 1 - p$. This is what p and q really mean.

	y_1	y_2	
x_1	a	$q - a$	q
x_2	$p - a$	$1 - p - q + a$	$1 - q$
	p	$1 - p$	

Now

$$\begin{aligned}\mathrm{E}(XY) &= a\,x_1y_1 + (q-a)\,x_1y_2 + (p-a)\,x_2y_1 + (1-p-q+a)\,x_2y_2,\\ \mathrm{E}(X)\mathrm{E}(Y) &= \big(qx_1 + (1-q)x_2\big)\big(py_1 + (1-p)y_2\big)\\ &= pq\,x_1y_1 + q(1-p)\,x_1y_2 + p(1-q)\,x_2y_1 + (1-p)(1-q)\,x_2y_2.\end{aligned}$$

Subtract coefficient by coefficient:

$$\begin{aligned}x_1y_1:&\quad a - pq,\\ x_1y_2:&\quad (q-a) - (q-pq) = -(a-pq),\\ x_2y_1:&\quad (p-a) - (p-pq) = -(a-pq),\\ x_2y_2:&\quad (1-p-q+a) - (1-p-q+pq) = a-pq.\end{aligned}$$

Hence

$$\mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y) = (a-pq)(x_1y_1 - x_1y_2 - x_2y_1 + x_2y_2) = (a-pq)(x_1-x_2)(y_1-y_2).$$

So if $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$, the left-hand side is zero and $(a-pq)(x_1-x_2)(y_1-y_2) = 0$, as required.

Independence

Any pair of random variables each taking two distinct values has a joint table of the form above, for suitable a, p, q (just name the cells). If $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$ then the product above is zero. Since $x_1 \neq x_2$ and $y_1 \neq y_2$, the last two factors are non-zero, so $a = pq$.

> **Ask yourself** We now know one cell factorises. Why does that settle the other three?

Because each other cell is a margin minus a cell we already know:

$$\begin{aligned}\mathrm{P}(X = x_1, Y = y_2) &= q - pq = q(1-p) = \mathrm{P}(X = x_1)\,\mathrm{P}(Y = y_2),\\ \mathrm{P}(X = x_2, Y = y_1) &= p - pq = p(1-q) = \mathrm{P}(X = x_2)\,\mathrm{P}(Y = y_1),\\ \mathrm{P}(X = x_2, Y = y_2) &= 1 - p - q + pq = (1-p)(1-q) = \mathrm{P}(X = x_2)\,\mathrm{P}(Y = y_2).\end{aligned}$$

Every cell is the product of its marginals, so X and Y are independent. (The converse, that independence gives $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$, is standard, so for two-valued variables the two conditions are equivalent.)

The three-valued example

Since A and B are each uniform on $\{-1, 0, 1\}$, we have $\mathrm{E}(A) = \mathrm{E}(B) = 0$, so we need a table with every row and column summing to $\frac{1}{3}$, with $\mathrm{E}(AB) = 0$ but not every cell

equal to $\frac{1}{9}$. A symmetric choice works:

	$B=-1$	$B=0$	$B=1$
$A=-1$	$\frac{1}{6}$	0	$\frac{1}{6}$
$A=0$	0	$\frac{1}{3}$	0
$A=1$	$\frac{1}{6}$	0	$\frac{1}{6}$

Each row and each column sums to $\frac{1}{3}$, so A and B each take $-1, 0, 1$ with probability $\frac{1}{3}$. The only non-zero products ab come from the corners:

$$\mathrm{E}(AB) = \tfrac{1}{6}(1) + \tfrac{1}{6}(-1) + \tfrac{1}{6}(-1) + \tfrac{1}{6}(1) = 0 = \mathrm{E}(A)\mathrm{E}(B).$$

But $\mathrm{P}(A=0, B=0) = \frac{1}{3} \neq \frac{1}{9} = \mathrm{P}(A=0)\,\mathrm{P}(B=0)$, so A and B are not independent. In words: $B=0$ exactly when $A=0$, and otherwise B is ± 1 with equal chances whatever A is.

Checks

- Try $X=Y$ with $x_1=y_1=1$, $x_2=y_2=0$: then $a=p=q$, and the identity says $\mathrm{Var}(X) = p - p^2$, which is right for a Bernoulli variable.
- The identity holds as a polynomial identity: we confirmed it exactly with rational arithmetic on 2000 random choices of x_i, y_i, a, p, q. It also checked the three-valued table: margins all $\frac{1}{3}$, $\mathrm{E}(AB)=0$, and $\mathrm{P}(A=0, B=0) = \frac{1}{3}$.

Common slips

- Showing $a = pq$ and stopping. Independence needs *every* cell to factorise, so the other three need checking (quickly).
- Forgetting to say why we may cancel $(x_1 - x_2)(y_1 - y_2)$: it is non-zero because the values are distinct.
- In the last part, offering $B = A^2$ or $B = |A|$. These are dependent with zero covariance, but B is never -1, so they don't have the required marginals.

Beyond STEP

The quantity $\mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y)$ is the covariance, and this question shows that "uncorrelated" is weaker than "independent", except for two-valued (indicator-like) variables, where the identity reduces everything to the covariance of two indicators. A related fact you will meet: for jointly normal random variables, zero covariance does imply independence. For more, see Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*.

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